USGLoc: Uncertainty-Aware Semantic Visual Localization from 3D Gaussian Splatting — Extended Report

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Abstract—This extended report accompanies the USGLoc letter. The letter presents an evidential localization framework on a 3D Gaussian Splatting (3DGS) map, built on one principle: the map's own uncertainty should drive every stage of localization and survive to the output. This report documents everything the letter's page budget excludes. We give the complete implementation and training recipes, an empirical analysis of the shared evidential head (EDL), statistics of the evidential map, a study of retrieval-encoder training on real data, per-scene calibration breakdowns, the full ablation grid, the robustness sweeps, per-stage runtime, and the DSLAM-protocol comparison with SGLoc referenced by the letter. Three implementation findings are reported that we believe transfer to other evidential-perception systems: (i) an EDL head's classification accuracy converges long before its uncertainty structure does; (ii) a shared head must be trained on its full deployment distribution — both input domains AND all scenes — or the unseen part's evidence collapses; and (iii) viewpoint submaps must be frustum-culled, or retrieval descriptors of a room-scale scene collapse onto a single vector.

Index Terms—Localization, semantic scene understanding, uncertainty quantification, evidential deep learning, Gaussian splatting.

I. Scope and Relation to the Letter

The RA-L letter (USGLoc: Uncertainty-Aware Semantic Visual Localization from 3D Gaussian Splatting) is the primary publication: it motivates the problem, derives the method (evidential map, vacuity-gated matching, vacuity-weighted refinement, tempered Laplace posterior), and reports the headline benchmark results. This report is the companion record. Section II fixes every implementation detail. Sections III–V analyze the three learned components with measured numbers. Sections VI–VIII hold the extended evaluation. Figure 1 of the letter (the pipeline overview) applies unchanged and is reproduced in Fig. 1.

This work was supported by JST SPRING, Japan Grant Number JPMJSP2102. (Corresponding authors: nakyoung@jaist.ac.jp, xiemhoang@vnu.edu.vn)

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TABLE I
Map-training configuration (identical for all scenes).

Stage	Parameter	Value
Init	frame / pixel stride	25 / 8
	voxel dedup / cap	2 cm / 200k
Fit	iterations @ resolution	7,000 @ 320×240
	frame stride / loss	10 / L1
	densify start–stop / reset	500–4,000 / 3,000
	lr (means*, quats, scales)	1.6×10^{-4}*, 10^{-3}, 5×10^{-3}
	lr (opacities, colors)	5×10^{-2}, 2.5×10^{-3}
Distill	frame stride / occlusion tol.	20 / 0.15 m
	CLIP / segmenter	ViT-B/16 laion2b / FastSAM-s
Opinions	full-evidence count n_0	4
Instances	voxel × class	0.4 m × argmax class

*scaled by the scene extent, as in 3DGS.

II. Implementation Details

A. Offline map training

Each scene is trained by a four-stage pipeline (scripts/train_map.py in the released code); Table I lists every hyperparameter. Gaussians are initialized by back-projecting the depth maps of every 25th mapping frame at a stride of 8 pixels, voxel-deduplicated at 2 cm, and capped at 200k seeds. The photometric fit uses gsplat rasterization [1] with DefaultStrategy densification, plain RGB colors (no spherical harmonics), and an L1 loss at 320×240 for 7,000 iterations, one random view per iteration over every 10th mapping frame. Semantic distillation runs FastSAM [2] masks with CLIP [3] ViT-B/16 crop embeddings on every 20th frame; each mask's embedding is accumulated onto the Gaussians whose centres project inside the mask, pass a depth-occlusion test ($|z-d| < 0.15$ m), and lie in front of the camera. On chess this yields 330,528 Gaussians, of which 94.1% receive at least one semantic observation from 9,348 mask embeddings; the full pipeline takes 94 s on an RTX 5070 Ti (16 GB).

B. Open-vocabulary class set

The evidential class count is C=25: wall, floor, ceiling, door, window, chair, sofa, table, desk, shelf, cabinet, monitor, keyboard, mouse, book, cup, bottle, plant, lamp, stairs, curtain, box, bag, kitchen appliance, sink. Class prototypes are CLIP text embeddings of "a photo of a ⟨class⟩ in a room". Pseudo-labels are the argmax cosine

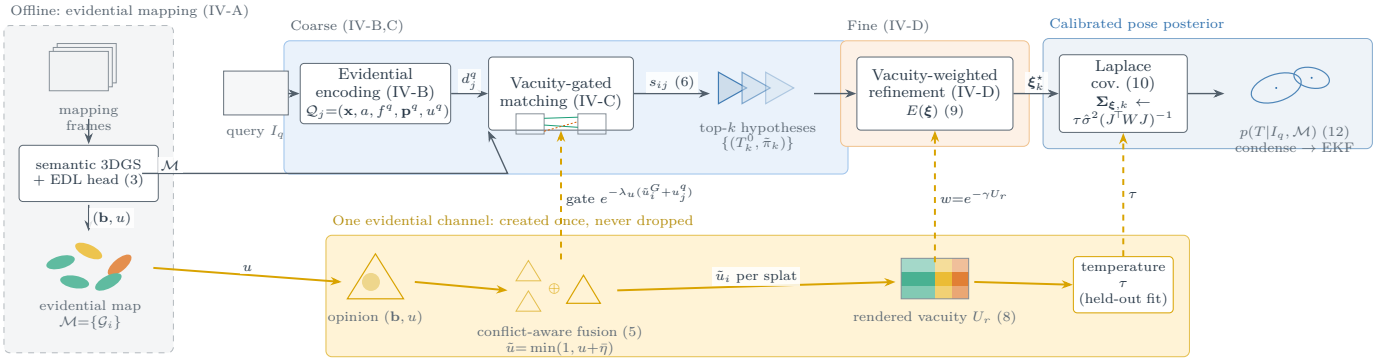


Fig. 1. The USGLoc pipeline (reproduced from the letter). Offline, an evidential semantic 3DGS map is trained; online, one evidential channel gates matching, weights rendering-based refinement, and calibrates the output posterior.

similarity, kept only when the top-1/top-2 margin exceeds 0.01.

III. The Shared Evidential Head

The head is a two-layer MLP ($512 \to 256 \to C$, softplus output, ∼138k parameters, 550 KB on disk) mapping CLIP features to Dirichlet evidence. It is trained once, then frozen and shared by the map and query sides. Neither FastSAM nor CLIP is fine-tuned.

A. Training data and schedule

The head needs no manually labeled dataset: its training set is generated automatically from the mapping frames themselves during map building, and Table II lists the full configuration. Two feature domains are pooled across ALL scenes (the reasons are Finding 2, Sec. III-C). Mask domain: every FastSAM mask crop embedded by CLIP on the distillation frames of every scene — 17,175 confidence-filtered embeddings over the seven 7Scenes environments (per scene 504–5,375). Gaussian domain: the per-Gaussian mask-averaged features of the already-mapped scenes, subsampled to at most 20k observed Gaussians per scene — 23,556 after filtering (chess + fire) — 40,731 training features in total. The trained head reaches 70–86% pseudo-label agreement on every scene's mask bank with the vacuity ordering intact. Labels are CLIP zero-shot pseudo-labels: cosine similarity against the C=25 text prototypes ("a photo of a ⟨class⟩ in a room"), argmax kept only when the top-1/top-2 margin exceeds 0.01; ambiguous features are deliberately excluded so the head assigns them little evidence at inference, which is the desired vacuity behavior. Optimization uses the EDL loss [4] (letter, Eq. 7): the Dirichlet Brier risk plus a KL regularizer toward the uniform Dirichlet on the misleading evidence, annealed as $\lambda_t = \min(1, \text{epoch}/10)$. Training takes seconds; the head is then frozen for all scenes and both sides of the pipeline. When a scene is mapped with a frozen head, stage 4 skips training entirely and only applies the head.

TABLE II
EDL head: complete training configuration (chess numbers).

Item	Value
architecture	MLP 512→256→25, ReLU, softplus
parameters / size	∼138k / 550 KB
training set	40,731 CLIP features (auto-generated)
mask domain	17,175 crops, ALL 7 scenes
Gaussian domain	23,556 sampled (mapped scenes)
labels	CLIP zero-shot vs. 25 prototypes, argmax
label filter	top-1/top-2 cosine margin > 0.01
loss	EDL: Brier risk + annealed KL [4]
KL anneal	$\lambda_t = \min(1, \text{epoch}/10)$
optimizer / lr / batch	Adam / 10^{-3} / 512
epochs	40 (Sec. III-B: why not 10)
wall-clock	seconds (RTX 5070 Ti)
reuse	trained once, frozen; shared everywhere

TABLE III
Mean vacuity u and top-class belief vs. training epochs. "Mix" = 50/50 mixtures of two class prototypes (the ambiguous regime); "junk" = random unit vectors.

	20 epochs			60 epochs		
	clean	mix	junk	clean	mix	junk
vacuity u	0.631	0.632	0.629	0.677	0.710	0.703
top belief $b_{\max}$	0.119	0.107	0.104	0.223	0.136	0.111

B. Finding 1: uncertainty structure needs more epochs than accuracy

On a controlled toy problem (4 orthonormal class directions in $\mathbb{R}^{16}$, 5% feature noise), classification accuracy exceeds 90% within ∼10 epochs, but the property the vacuity gate relies on — ambiguous inputs earn more vacuity than clean ones — is absent at 20 epochs and only emerges by 60 (Table III). We therefore train for 40 epochs by default and regression-test the ordering.

C. Finding 2: the shared head must train on both input domains

The head is applied to raw mask-crop CLIP features (query side) and to per-Gaussian features averaged over many overlapping masks (map side). Averaging shifts features off the raw-mask manifold. A head trained only on mask features assigns the averages near-zero evidence:

TABLE IV
Chess map vacuity with a mask-only vs. dual-domain head. u_{eff} is the fused primitive-level effective vacuity (letter, Sec. IV-A); q_p denotes quantiles.

	mask-only head	dual-domain head
per-Gaussian median u	0.997	0.761
per-Gaussian q_{10}	0.995	0.223
primitive u_{eff} $q_{10}/q_{25}/q_{50}$	0.82 / 0.89 / 0.99	0.001 / 0.26 / 0.97
primitives with $u_{\text{eff}}<0.7$	8 / 1916	629 / 1916
primitives with $u_{\text{eff}}<0.1$	0 / 1916	402 / 1916

TABLE V
Per-scene evidential map statistics (SfM pseudo-GT poses).

Scene	splats	prims	obs.	masks	med u	s
chess	330,528	1,916	94.1%	9,348	0.761	94
fire	–	–	–	–	–	–
heads	–	–	–	–	–	–
office	–	–	–	–	–	–
pumpkin	–	–	–	–	–	–
redkitchen	–	–	–	–	–	–
stairs	–	–	–	–	–	–

on chess the median per-Gaussian vacuity was 0.997 despite 94% observation coverage, and the primitive-level spread collapsed (Table IV) — the gate had nothing to discriminate. Training the same head on the union of both domains (each pseudo-labeled identically) restores a full spread at no architectural cost.

The same lesson holds across scenes: a dual-domain head trained on chess alone reached only 8.9% pseudo-label agreement on fire's features, and fire's map evidence collapsed exactly as above (median vacuity 0.994). Pooling the training set over all scenes (Sec. III-A) restores 70–86% agreement everywhere. We regard this as a general caution for shared evidential heads: share the weights, and share the full training distribution — every input domain and every environment the head will score.

D. Evidence scaling and fusion behavior

Per-Gaussian evidence is scaled by observation count, $e \leftarrow e \cdot \min(1, n/n_0)$ with n_0=4: a never-observed Gaussian has u=1 exactly, and evidence accumulates with support, as subjective logic prescribes. Note that with C=25 classes the per-observation vacuity floor is high ($u = C/S$), which is not a defect: cluster-level DST fusion multiplies member vacuities, so agreement across members compounds into confidence (the bimodal primitive distribution of Table IV), while disagreement routes conflict into vacuity under Yager's rule [5], [6]. The absolute scale is absorbed by the gate sharpness λ_u and pixel-weight sharpness γ; only the ordering matters, which Sec. III-B verifies.

IV. Evidential Map Statistics

Table V summarizes the per-scene maps. [TODO: fill the remaining six scenes after the batch build (plan step 3.1)]

V. Retrieval-Encoder Training on Real Data

Positive InfoNCE pairs are (FastSAM+CLIP+EDL instances of frames at offsets $\{0, \pm4, \pm8\}$ around a submap anchor, the submap's reference primitives); anchors are every 20th mapping frame, and the neighboring frames act as natural augmentation of the query side.

A. Finding 3: submaps must be frustum-culled

With radius-only submaps (r=3 m) in a room smaller than the radius, every chess submap contained 768–1,529 of the 1,916 primitives; all reference scene descriptors collapsed onto one vector (pairwise cosine 0.9996–1.0000) and the symmetric InfoNCE loss sat exactly at chance, $\ln(24) \approx 3.178$, for 400 iterations. Culling each submap to the primitives that project into its anchor view (in front of the camera, inside the image) yields 154–340 primitives per submap with Jaccard overlap ∼0.40 between neighboring anchors, and the loss [TODO: report the converged InfoNCE value and a small loss-curve figure]. This mirrors the covisibility submaps of SGLoc [7]: viewpoint content, not a distance ball, is the retrieval unit.

VI. Extended Evaluation Protocol

All headline numbers follow the letter's protocol; this section fixes the details a reproduction needs. Ground truth: the SfM pseudo-GT poses of Brachmann et al. [8] for 7Scenes [9] and per-scene SfM focal lengths (525.14–527.75); the DSLAM and SfM references differ by ∼1 cm on the same frame, larger than the gaps separating recent methods, so mixing them is not benign. Hyperparameter tuning: on a validation split carved from train-side frames, disjoint from map-fitting and distillation frames; the test split is evaluated once with frozen settings. Temperature: τ is fitted on train-side calibration frames, never on test queries, and the full test split is reported. Seeds: the Ours and ablation rows are the median of 3 seeds. [TODO: state the exact validation frame rule and calibration frame count once frozen (plan steps 2.6 and 3.2)]

VII. Extended Results

A. Per-scene calibration

TABLE VI
Per-scene coverage at nominal 68.3/95.4/99.7% (tempered), AUSE and NLL. The letter reports the aggregate; ambiguous scenes (stairs) probe the Gaussian assumption per hypothesis.

Scene	cov. 68.3	cov. 95.4	cov. 99.7	AUSE$_t$	NLL
chess	–	–	–	–	–
fire	–	–	–	–	–
heads	–	–	–	–	–
office	–	–	–	–	–
pumpkin	–	–	–	–	–
redkitchen	–	–	–	–	–
stairs	–	–	–	–	–

B. Full ablation grid

[TODO: full grid over: EDL vs softmax head; gate off (λ_u=0); pixel weight off (γ=0); Dempster vs Yager; K=1 vs 3; raw vs tempered; fixed-scaled covariance baseline — with bootstrap CIs (plan steps 3.3 and 3.6)]

C. Robustness sweeps

[TODO: error AND coverage/NLL vs (a) mapping budget (fit stride 10/40/80), (b) distillation sparsity (stride 20/80/200), (c) degraded semantics; the claim under test: accuracy degrades gracefully while calibration holds (plan step 3.4)]

D. Per-stage runtime

[TODO: retrieval / refinement / Laplace+condensation ms on RTX 5070 Ti (plan step 3.5); map training is 94 s/scene (Sec. II)]

VIII. DSLAM-Protocol Comparison with SGLoc

SGLoc [7] evaluates on 4 of 7 scenes under the original DSLAM ground truth, which the pseudo-GT study [8] shows to be the easier convention; the letter therefore keeps this comparison out of Table I. For completeness: [TODO: run USGLoc under DSLAM GT (omit –sfm-pose-dir) on chess, fire, heads, office and tabulate against SGLoc's reported numbers]

TABLE VII
DSLAM-protocol comparison (median cm / deg).

	chess	fire	heads	office
SGLoc	–	–	–	–
USGLoc (ours)	–	–	–	–

IX. Additional Figures

[TODO: Fig.: query RGB / rendered vacuity U_r / pixel weight w triptychs on 2–3 queries (plan step 4.2)]

[TODO: Fig.: per-query pose-error CDFs vs the strongest baselines]

[TODO: Fig.: refinement error vs iteration, with and without the vacuity pixel weight]

Reproducibility

All numbers in this report regenerate from the released code: train_map.py (Sec. II–IV), train_encoders.py (Sec. V), and run_7scenes.py (Sec. VII); the per-table commands are listed in the repository. [TODO: insert repository URL at camera-ready; the anonymized mirror during review]

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